Danielle Grau PhD

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Education

Georgia Institute of Technology Aug 2021 - Jun 2026
PhD Candidate in Earth & Atmospheric Sciences GPA 3.87

Florida International University Jan 2018 - May 2021
B.S. in Physics (*Summa Cum Laude*) GPA 3.92

Relevant Coursework: Electromagnetism 1 & 2, Mathematical Methods in Physics, Modern Physics, Advanced Modern Physics, Radiation Detection, Intermediate & Advanced Lab, Quantum Mechanics 1 & 2, Classical Mechanics 1, Thermodynamics, Physical Climatology, Environmental and Exploration Geophysics, Land Remote Sensing, Earth System Modeling, Physical Hydrology, Advanced Environmental Analysis, Fluid Dynamics & Synoptic Meteorology, Glacier & Ice Sheet Dynamics, Carbon Dioxide Removal, Machine Learning in Environmental Systems

Publications

1. **Grau, D.**, M. Poinelli, N.J. Schlegel, H. Seroussi, A.A. Robel. Accelerated Ice Loss from Antarctica in Simulations with Calving Driven by Supraglacial Melt Lake Depth. (Submitted *Geophysical Research Letters*, June 2026).
2. **Grau, D.**, A. Hussain, A.A. Robel (2025). Predicting Mean Depth and Area Fraction of Antarctic Supraglacial Melt Lakes with Physics-Based Parameterizations, *Nature Communications*, In Press.

Research and Professional Experience

Graduate Research Assistant
Georgia Institute of Technology Aug 2021 - Jul 2026

- **Analyze Observational, Remote-Sensing Datasets:** Developed MATLAB and Python-based workflows to analyze the statistical surface roughness properties of the entire Antarctic and Greenland Ice Sheets from NASA ICESat-2 Satellite Altimetry tracks.
- **Designed and Parallelized Monte-Carlo Simulations** Developed a MATLAB-based workflow that generates a random, statistically rough surface and uses a cellular automaton model to simulate surface water distribution and routing.
- **Extrapolate Relationships from Large Simulated Datasets**Utilized MATLAB's Curve Analysis library to derive mathematical relationships from a Monte-Carlo simulated data set.
- **Model Configuring and Tuning:** Performed Transient Projections for Antarctic Ice Sheet Loss from 2015 to 2300 with external GCM and Ocean Forcing.
- **Large Ice-Sheet Model Component Development:** Developed new class and calving scheme within NASA's Ice-Sheet and Sea-Level System Model (ISSM) using MATLAB and C++.
- **Present Scientific Research:** Present academic research in both poster and oral formats at national and international conferences.
- **Scientific Writing:** Composed and published scientific articles in high-profile journals (*Nature Communications*)
- **Collaboration**Collaborated with other professional scientific researchers from other research institutions.
- **Mentor Undergraduate Researchers** Provide resources and guidance on research projects for three undergraduate student researchers.
- **Secure over $50,000 in fellowships and grants:** Internal fellowships and the NASA Georgia Space Grant Consortium.

Undergraduate Researcher
Georgia Institute of Technology May 2020 - May 2021

- **Signal and Power Spectral Density Analysis:** Developed a Python-based workflow to perform PSD analysis on ICESat-2 land-ice Altimetry tracks.

Universidad de Magallanes Nov. 2023
Patagonian Ice-field Research Program (PIRP) Student Researcher
Grey Glacier, Chile

- Measured Calving Events at Grey Glacier, Chile using an Aquarian Audio & Scientific H2d Hydrophone faster.

- Data was analyzed using signal processing in the field. Tech: MATLAB, Signal Processing.

University of Washington Aug. 2023
ICESat-2 Hack-week Researcher, *Seattle, Washington*
- Develop a Python-based workflow to analyze grounding zone features in Antarctica using NASA's ICESat-2 Land Ice Altimetry.
- Tech: CryoCloud, SlideRule Earth

Honors and Recognitions

- EAS Graduate Student Service Award 2025
- Georgia Tech Pathbreakers Fellow 2023-Present
- Presidential Fellowship 2021-Present
- Florida International University CASE Service Award April 2021
- Florida International University CASE Outstanding Academic Achievement (Undergraduate) April 2021
- Fred Hoover Scholar 2020-2021
- Dean's List 2018 - 2021
- Florida Bright Future's Scholar 2018 - 2021

Skills

Coding Languages: Python, MATLAB, C++, Fortran, LaTeX, Mathematica **Frameworks:** Scikit, Pandas, Tensorflow, Keras, PBS, Slurm
Operative Systems: Macintosh, Linux Shell Environment and Scripting
Computation Tools: HPC, Parallel Computing, Machine Learning, Cloud Computing, MS Office, Adobe Illustrator, QGIS
Languages: English, Spanish, French
Soft Skills: Leadership, Event Organization and Management, Scientific and Professional Writing, Public Speaking
Instrumentation: Instrument Experience with Hydrophone, Seismometers, Ablation Stakes, Tree Core Sampler, and Ice Steam Drill
Field Experience: River-Crossing, Traversing Glacier and Alpine Terrain, Ice Climbing, and Leave No-Trace Camping Techniques

Conference Presentations

1. "*Modeling interaction between supraglacial melt lakes and calving in transient Antarctic simulations*", European Geophysical Union Assembly 2026, Oral Presenter. Vienna, Austria. May 2026.
2. "*Modeling interaction between supraglacial melt lakes and calving in transient Antarctic simulations*", American Geophysical Union Meeting 2025, Oral Presenter. New Orleans, LA, USA. Dec 2025
3. "*A Physics-Based Parameterization of Mean Melt Lake Depth and Area Fraction of Supraglacial Melt Lakes*", European Geophysical Union Assembly 2025, Poster Presenter. Vienna, Austria, May 2025.
4. "*A Physics-Based Parameterization to Predict Mean Depth & Areal Coverage of Supraglacial Melt Ponds*", West Antarctic Ice Sheet Work Shop, Poster Presenter. Cloquet, MN, USA. Sep 2023.
5. "*A Physics-Based Parameterization to Predict Mean Depth and Areal Coverage of Supraglacial Melt Ponds*", Climate Sustainability Challenges and Opportunities Workshop, Oral Presenter. Atlanta, GA, USA. Aug 2023.
6. "*A Statistical Parameterization of Supraglacial Melt Pond Are and Depth on Fractal Ice Sheet Surfaces from Percolation Theory* (1160496)", American Geophysical Union Annual Meeting 2022, Poster Presenter. Chicago, IL, USA. Dec 2022.
7. "*Statistical Models of Supraglacial Melt Ponds Characteristics from Monte-Carlo Simulations*", Women in High-Performance Computing, Poster Presenter. Atlanta, GA, USA. April 2022.

Leadership Experience & Community Engagement

Graduate Student Senator
Georgia Institute of Technology Aug 2025 - Apr 2026
- **Review and Vote on Legislation:** Review and Vote on incoming bills from various committees.

President of Graduate Students in Earth & Atmospheric Sciences
Georgia Institute of Technology Jun 2024 - May 2025

- **Delegation:** Collaborate with other members of the executive board and assign tasks.
- **Organize Events:** Graduate Student Socials, Professional Development Events, Department-wide events.
- **Coordinate with Department Faculty and Staff:** Deliver concerns and worries from graduate students and collaborate on solutions.

APS IDEA Member
Florida International University Aug 2020 - Apr 2021

- **Undergraduate Representative of Florida International University's APS IDEA team** Attended monthly meetings to discuss how to better diversity, equity and inclusion for stakeholders in the Physics discipline across the country

ΣΠΣ **President (FIU Chapter)**
Florida International University May 2020 - Apr 2021

- **Delegate:** Delegate tasks to executive board members throughout the academic year of 2020- 2021.
- **Design and Host Events:** Designed, planned, and executive events
- **Designed Professional Development Workshops:** Designed, Wrote, and Hosted workshops for developing CV/Resume, Cover Letters, Introductory Python, and Introductory LaTeX
- **Collaborate with academic departments and student organizations:** Collaborated with other academic departments to host

ΣΠΣ **Vice President (FIU Chapter)**
Florida International University May 2019 - Apr 2020

Managed Mentorship Program: Oversaw the Pion Mentorship Program, which matches lower-division undergraduate students with senior undergraduate seniors

Coordination: Coordinated with Physics Department Staff and Faculty to send undergraduate students to national and regional conferences such as APS March and the South East American Section of the American Physical Society (APS).

ΣΠΣ **Secretary (FIU Chapter)**
Florida International University Sep 2018 - Apr 2019

Marketing and Communication: Oversaw communications between ΣΠΣ members and executive board to advertise events, workshops, and opportunities

Record Keeping of Meeting Minutes and Attendance: Kept record of attendance to events, weekly tutoring, workshops, and meetings as well as kept minutes for organizational meetings.

Teaching and Mentoring Experience

Georgia Institute of Technology
Mentor for Undergraduate Mentor
Teaching Assistant Aug 2022 - Dec 2022

- Hosted Hybrid Office Hours twice a week
- Created Answer Keys for & Graded Assignments
- Corresponded with Students on Course Materials & Assignments

Florida International University|Department of Physics
Learning Assistant Jan 2020- May 2021

- Aided students with materials for introductory Physics Labs
- Assisted in Proctoring Exams
- Hosted Review & Help Sessions

Professional Memberships & Roles

European Geophysical Union	2025- Present
GeoLatinas	2022-Present
International Glaciological Society	2023-Present
American Geophysical Union	2022-Present
Phi Beta Kappa	2021-Present
National Physics Honor Society (ΣΠΣ)	2020-Present
National Society of Leadership and Success	2019 - Present
Golden Key International Honour Society	2018-Present
Society of Physics Students	2019-Present
Sigma Alpha Lambda	2018 - Present